



Volume and scale

Teacher resource

Question 1

A rectangular prism is 2 cm wide, 4 cm long and 3 cm high.
The rectangular prism is enlarged.
After enlargement the prism has a volume of 192 cm^3 .

What scale factor is used to enlarge the rectangular prism?

2

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4

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8

☐

12

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Answer: A

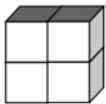
Skill: Students understand effect of scale factor on volume

Additional questions

1. A rectangular prism is 4 cm long, 3 cm wide and 2 cm high. After it is enlarged it is 32 cm long, 24 cm wide and 16 cm high. What scale factor is used to enlarge the prism?
2. How will the volumes of the two prisms in Question 1 compare?

Question 2

Ned made this model with four cubes.



How many more cubes are required to enlarge this model by a scale factor of 3?

_____ cubes

Answer: 104 cubes

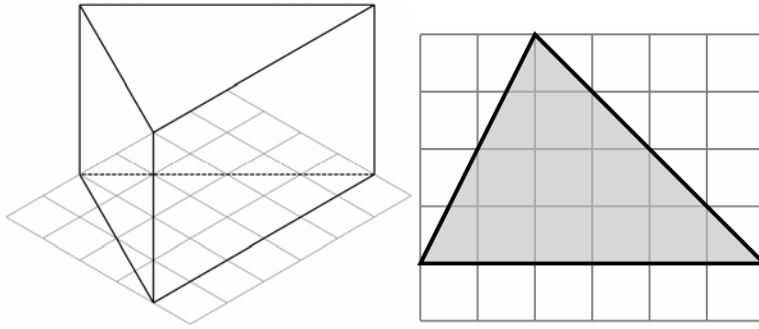
Skill: Students apply links between scale factors and volume.

Additional questions

1. What are the dimensions of the enlarged model?
2. The dimensions of the model are increased by a factor of 3. By what factor will the volume increase?
3. By what factor will the surface area increase?

Question 3

The diagrams show two views of a prism on a grid.



Each square on the grid has an area of one square centimetre.
The lengths of the vertical edges of the prism are 4 centimetres.

What is the volume of the prism in cubic centimetres?

_____ cubic centimetres

Answer: 48 cm³

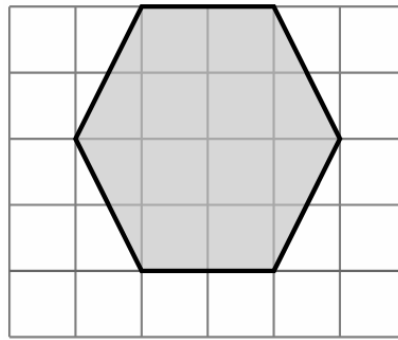
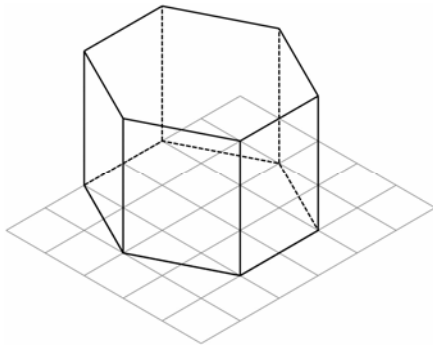
Skill: Students calculate volume of a prism.

Additional questions

1. What would happen to the volume of the prism if the length of the vertical edges was doubled?
2. What would happen to the area of the triangular base of the prism if the length of each square on the grid was doubled?
3. What would happen to the volume of the prism if all of its dimensions were increased by a factor of 2?

Question 4

The diagrams show two views of a prism on a grid.



Each square on the grid has an area of four square centimetres.
The volume of the prism is 312 cubic centimetres.

What is the perpendicular height of the prism in centimetres?

_____ centimetres

Answer: 6.5 cm

Skill: Students understand the relationship between the area of the base and the volume of a prism.

Additional questions

1. How would the volume of the prism change if all its dimensions were reduced by a factor of five (in other words, each dimension becomes a fifth of the original length)?
2. How would the surface area of the prism change if all its dimensions were reduced by a factor of five?

Student activity: Volume and scale

Question 1

A rectangular prism is 2 cm wide, 4 cm long and 3 cm high.
The rectangular prism is enlarged.
After enlargement the prism has a volume of 192cm^3 .

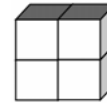
What scale factor is used to enlarge the rectangular prism?

2 4 8 12

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Question 2

Ned made this model with four cubes.

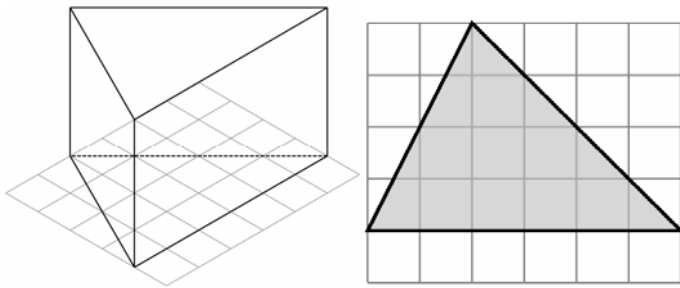


How many more cubes are required to enlarge this model by a scale factor of 3?

_____ cubes

Question 3

The diagrams show two views of a prism on a grid.



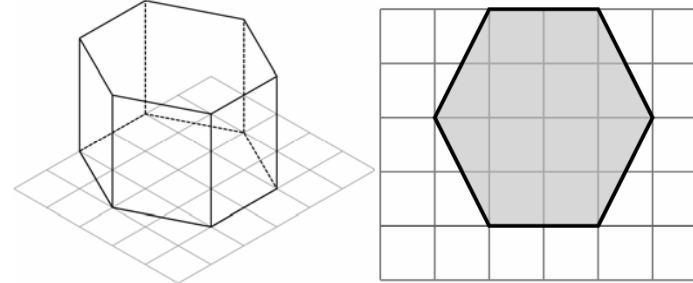
Each square on the grid has an area of one square centimetre.
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What is the volume of the prism in cubic centimetres?

_____ cubic centimetres

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